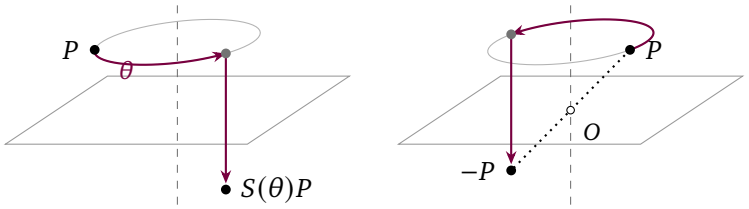
Symmetries in Physics — Exercise Sheet 7

Molecular vibrations

Variant: Questions only

Exercise 1. The table in Lecture 7 lists χ_{vec} for the rotations C_2, C_3, C_4, C_6, the mirror plane $\sigma = S(0)$ and the inversion $i = S(\pi)$. It omits the rotoreflections $S_n = S(2\pi/n)$. A rotoreflection is a rotation through $2\pi/n$ followed by the reflection in the plane perpendicular to its axis (left picture).



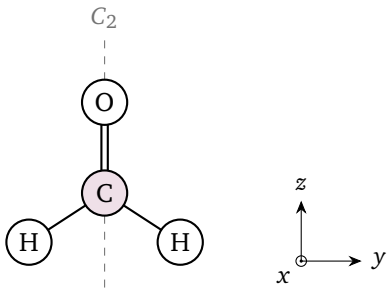
(a) $S(\theta)$ (b) S_2

(i) Compute $\chi_{\text{vec}}(S_n)$ for $n = 3, 4, 6$.

(ii) Show that S_2 is the inversion (right picture).

(iii) The lemma of the lecture says that a rotoreflection through a nonzero angle leaves only the centre in place. Prove this for S_n with $n \geq 3$. Which values can N_g then take?

Exercise 2. Formaldehyde H_2CO is planar. The C–O bond bisects the H–C–H angle. Place it as water was placed in Lecture 7: in the yz-plane, with the C–O bond along the z axis. The molecular plane is the page.



(i) Find the point group with the three questions of Lecture 7: the principal axis, the vertical planes, a horizontal plane. Name the planes $\sigma_v(yz)$ and $\sigma'_v(xz)$, as for water.

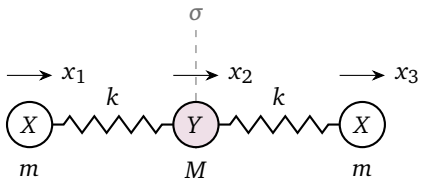
(ii) Using the lemma on atoms left in place, compute χ_{disp} and χ_{vib}.

(iii) Decompose Γ_{vib} with the C_{2v} character table of Lecture 7's water tutorial. With the planes named this way, it reads

C_{2v}	E	C_2	$\sigma_v(yz)$	$\sigma'_v(xz)$	
A_1	1	1	1	1	z
A_2	1	1	−1	−1	
B_1	1	−1	1	−1	y
B_2	1	−1	−1	1	x

(iv) A nonlinear planar molecule has $2N-3$ in-plane and $N-3$ out-of-plane vibrations. Check your answer against these counts, using the characters of the molecular plane $\sigma_v(yz)$.

Exercise 3. A linear triatomic molecule X–Y–X has masses m, M, m and springs of stiffness k between neighbours. The atoms move along the molecular axis only, and their displacements x_1, x_2, x_3 are all measured in the same direction. The mirror σ through Y, perpendicular to the axis, exchanges the end atoms. The point group of a linear molecule is infinite, so, as in Lecture 7, we do not use the whole group. We use only the group $\{E, \sigma\}$.

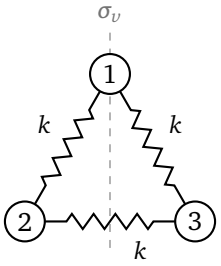


(i) Using the displacement representation of Lecture 7, show that $\rho(\sigma)(x_1, x_2, x_3) = (-x_3, -x_2, -x_1)$, and that $\rho(\sigma)$ commutes with the equations of motion.

(ii) Split $\mathbb{R}^3$ into the even and the odd subspace of $\rho(\sigma)$, block-diagonalise the equations of motion, and find the vibration frequencies.

(iii) Which mode is the symmetric stretch, and which the antisymmetric stretch? The plain exchange $P\colon (x_1, x_2, x_3) \mapsto (x_3, x_2, x_1)$ also commutes with the dynamics. Why does it give the opposite labels?

Exercise 4. Three equal masses m sit at the corners of an equilateral triangle, and each pair is joined by a spring of stiffness k. The springs are unstretched at equilibrium. The masses move only in the plane of the triangle. The symmetry group is $C_{3v} \cong S_3$: the rotations through $2\pi/3$ and the three mirror lines through the corners, as in Lecture 5.



(i) Adapt the lemma on atoms left in place to the plane: replace χ_{vec} by the trace of the 2×2 matrix $R(g)$ of each operation. Show that $\chi_{\text{disp}} = (6, 0, 0)$ on the classes $(E,\ 2C_3,\ 3\sigma_v)$, and that $\Gamma_{\text{disp}} = A_1 \oplus A_2 \oplus 2E$.

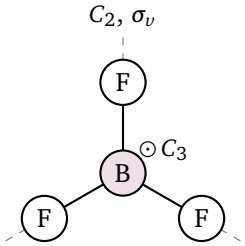
(ii) The two translations transform as (x, y), that is, as E. The rotation in the plane is labelled by an angle, which the operation g multiplies by $\det R(g)$. Show that the rotation transforms as A_2, and deduce $\Gamma_{\text{vib}} = A_1 \oplus E$.

(iii) The type A_1 occurs once, so its symmetry-adapted vector is a normal mode. In it every mass moves radially outward by the same distance s (the *breathing mode*). Show that each spring lengthens by $\sqrt{3}\,s$, and find ω^2.

(iv) The type E occurs twice in Γ_{disp}, so by Lecture 6's block theorem the dynamical matrix $D = K/m$ acts on the isotypic component of E through a 2×2 reduced matrix $F^{(E)}$.

One of its eigenvalues is 0, because the translations lie in this component. Show that $\operatorname{tr} D = \sum_i n_i \operatorname{tr} F^{(i)}$, and use it to find the other eigenvalue, the squared frequency ω_E^2 of the E doublet, without computing $F^{(E)}$.

Exercise 5 (boron trifluoride, end to end). Run the full vibrational analysis of planar BF_3. Its point group is D_{3h}. A B atom sits at the centre, and three F atoms sit at the corners of an equilateral triangle. The page is the molecular plane σ_h. The C_3 axis is perpendicular to it through B. Each C_2 axis runs along a B–F bond. Each mirror σ_v contains the C_3 axis and one C_2 axis.



(i) Write down χ_{disp} from the lemma on atoms left in place (Lecture 7).

(ii) Reduce it, subtract translations and rotations, and obtain Γ_{vib}.

(iii) Two types occur once in Γ_{vib}, so their modes are fixed by symmetry alone. Draw them.

Data. D_{3h} has classes E, $2C_3$, $3C_2$, σ_h, $2S_3$, $3\sigma_v$, of sizes $1, 2, 3, 1, 2, 3$. Here the *operation* S_3 (not the group) is the rotoreflection: rotation by $2\pi/3$ followed by reflection in σ_h. The complete character table is

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
A_1'	1	1	1	1	1	1
A_2'	1	1	−1	1	1	−1
E'	2	−1	0	2	−1	0
A_1''	1	1	1	−1	−1	−1
A_2''	1	1	−1	−1	−1	1
E''	2	−1	0	−2	1	0

The class sizes are the coefficients in the headings. One prime marks a type even under σ_h, and two primes a type odd under it. Translations transform as $z \sim A_2''$, $(x, y) \sim E'$, and rotations as $R_z \sim A_2'$, $(R_x, R_y) \sim E''$.